Due on Friday April 29th

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1. The four-vector A^μ is found to be timelike in one inertial frame.

   (a) **[4 pts]** Write an equation or inequality expressing the fact that A^μ is timelike.

   (b) **[3 pts]** Explain why A^μ has to be timelike in any inertial frame.

2. The kinetic energy of motion of a particle is the relativistic total energy minus the rest energy.

   (a) **[5 pts]** A particle has rest mass M and speed v. If $v \ll c$, then show that the kinetic energy due to motion is approximated by the well-known non-relativistic expression for the kinetic energy.

   (b) **[5 pts]** Derive the leading correction to the non-relativistic expression.

   (c) **[5 pts]** Express the momentum p of the particle as a function of the mass M and the kinetic energy of motion, which we will call T. Your expression should contain M and T, not v or γ_v.
   Hint: use the relation we referred to as Very Important in class.

3. **[7 pts]** A π-meson of mass M at rest disintegrates into a μ-meson of mass m_μ and a neutrino of effectively zero mass. Show that the kinetic energy of motion of the μ-meson is

$$T = \frac{(M - m_\mu)^2}{2M} c^2$$

   As usual, for solving such problems you first draw the situations before and after the reaction/decay, write down the equations for energy conservation and for momentum conservation in terms of the most useful variables, and then manipulate these equations as required.

   You should decide early whether it is helpful to work with speed and γ, or whether to avoid these and work with momentum instead.

   You probably want to use $c = 1$ units for this problem. If you do, you will not get the c^2 factor above, but I don't mind.

4. (a) [**4 pts**] We consider the Lorentz transformation

$$x'^{\mu} \;=\; \Lambda^{\mu}_{\nu} x^{\mu}$$

between frames S and S', which are coincident at time $t = t' = 0$. Frame S' moves with velocity u with respect to frame S, along the x^1 (or x'^1) direction. Write down the 4×4 matrix representing the Lorentz transformation Λ^{μ}_{ν}.

(In the early part of the module, we generally ignored the x^2 and x^3 directions and wrote the transformation as a 2×2 matrix.)

If you use the symbol γ, do indicate which speed it corresponds to.

(b) [**4 pts**] Now consider the Lorentz boost from S to a frame S'', coincident at time $t = t'' = 0$, that moves with velocity v with respect to frame S, but along the x^2 or x''^2 (i.e., the y or y'') direction. Write down the 4×4 matrix representing this Lorentz transformation.

5. [**4 pts**] The mass m and the charge q of a particle are four-scalars. Explain why the combination

$$(m, q, m, q)$$

is not a four-vector.

6. As measured in frame S, the coordinates of event A are (ct_A, x_A) and the coordinates of event B are (ct_B, x_B). The interval between the events are spacelike.

There is then a frame S' in which the two events are simultaneous.

(a) [**4 pts**] Plot the two events on a spacetime diagram, taking the S frame to have ct and x axes perpendicular to each other. Plot also the axes of the S' frame, so that it is clear that A and B are simultaneous in this frame. You only get credits if this last feature is evident in your drawing.

(b) [**5 pts**] Calculate the speed v of the S' frame with respect to the S frame, in terms of t_A, t_B, c, x_A and x_B. You could either use your drawing, or use the Lorentz transformations. (Either is fine.)